

TPA-2

HYBRIDISATION GEOMETRY, SHAPE AND BOND ANGLES IN MOLECULES

- Identify the correct match.

(i) XeF_2	(a) Central atom has sp^3 hybridisation and bent geometry.
(ii) N_3^-	(b) Central atom has sp^3d^2 hybridisation and octahedral.
(iii) PCl_6^- (PCl_5 (s) anion)	(c) Central atom has sp hybridisation and linear geometry.
(iv) ICl_2^+ (I_2Cl_6 (l) cation)	(d) Central atom has sp^3d hybridisation and linear geometry.
(A) (i – a), (ii – b), (iii – c), (iv – d)	(B) (i – d), (ii – b), (iii – d), (iv – c)
(C) (i – b), (ii – c), (iii – a), (iv – d)	(D) (i – d), (ii – c), (iii – b), (iv – a)
- Which of the following statement is true for IO_2F_2^- ?

(A) The electrons are located at the corners of a trigonal bipyramidal but one of the equatorial pairs is unshared.

(B) It has sp^3d hybridisation and is T-shaped.

(C) Its structure is analogous to SF_4 .

(D) (A) and (C) both
- Which reaction involves a change in the electron-pair geometry for the under lined element ?

(A) $\text{BF}_3 + \text{F}^- \longrightarrow \text{BF}_4^-$	(B) $\text{NH}_3 + \text{H}^+ \longrightarrow \text{NH}_4^+$
(C) $2\text{SO}_2 + \text{O}_2 \longrightarrow 2\text{SO}_3$	(D) $\text{H}_2\text{O} + \text{H}^+ \longrightarrow \text{H}_3\text{O}^+$
- The ONO angle is maximum in :

(A) HNO_3	(B) NO_2^+	(C) HNO_2	(D) NO_2
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- Which statement is correct for N_3^- ion

(A) It is bent molecule	(B) Bond angle is $< 120^\circ$
(C) Central atom is sp^2 hybridized	(D) None of these
- In which of the following bond angle is maximum

(A) NH_3	(B) NH_4^+	(C) PCl_3	(D) SCl_2
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- Consider the following statement and arrange in the order of true / false.

S_1 : In SnCl_2 the bonding takes place in ground state and the bond angle Cl-Sn-Cl is slightly less than 120° .

S_2 : The molecular geometry of XeF_7^+ is pentagonal bipyramidal having two different Xe-F bond lengths.

S_3 : In SF_4 , the bond angles, instead of being 90° and 180° are 89° and 177° respectively due to the presence of a lone pair.

(A) T T T	(B) F T T	(C) T T F	(D) T F T
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- Consider the following statements
 - Steric number '7' gives ' sp^3d^3 ' hybridisation.
 - In ClF_3 at least one bond angle is exactly 180°
 - Lone pair does not cause any distortion in the bond angle.

The above statements 1, 2, 3 respectively are (T = True, F = False)

(A) T F F	(B) T T F	(C) F T F	(D) T T T
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- Which one has highest bond angle.

(A) NH_3	(B) PH_3	(C) H_2O	(D) CH_4
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- Statement-1 : The bond angles of BrNO , ClNO and FNO are approximately 114.5° , 113.3° and 110.1° respectively.
 Statement-2 : The hybridisation of central N atom in all three compounds is sp^2 .

(A) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is correct explanation for STATEMENT-1

(B) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is not correct explanation for STATEMENT-1

(C) STATEMENT-1 is true, STATEMENT-2 is false

(D) STATEMENT-1 is false, STATEMENT-2 is true

(E) Both STATEMENTS are false

11. Statement-1 : Molecules having different hybridisation can have same shape.
Statement-2 : The shape of a molecule does not depend on the hybridisation but it depends on the energy factor.
(A) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is not correct explanation for STATEMENT-1
(C) STATEMENT-1 is true, STATEMENT-2 is false
(D) STATEMENT-1 is false, STATEMENT-2 is true
(E) Both STATEMENTS are false
12. Statement-1 : SO_2 , NO_3^- and CO_3^{2-} are isoelectronic as well as isostructural species.
Statement-2 : The d and f-orbitals do not shield the nuclear charge very effectively. Therefore, there is significant reduction in the size of the ions, just after d or f orbitals have been completely filled.
(A) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is not correct explanation for STATEMENT-1
(C) STATEMENT-1 is true, STATEMENT-2 is false
(D) STATEMENT-1 is false, STATEMENT-2 is true
(E) Both STATEMENTS are false
13. The POCl_3 molecule has the shape of an irregular tetrahedron with the P atom located centrally. The Cl-P-Cl angle is found to be 103.5° . Give a qualitative explanation for the deviation of this structure from a regular tetrahedron.
14. Arrange the following in the increasing order of bond angles :
(i) NO_2^+ , BF_4^- , CO_3^{2-} (ii) SO_2Cl_2 , SO_3 , SF_6 (iii) OCl_2 , OF_2 , OBr_2
(iv) Equatorial bond angle in PCl_5 , XeO_2F_2 , PBr_5
15. Arrange the following in order of decreasing N – O bond length : NO_2^+ , NO_2^- , NO_3^-
(A) $\text{NO}_3^- > \text{NO}_2^+ > \text{NO}_2^-$ (B) $\text{NO}_3^- > \text{NO}_2^- > \text{NO}_2^+$
(C) $\text{NO}_2^+ > \text{NO}_3^- > \text{NO}_2^-$ (D) $\text{NO}_2^- > \text{NO}_3^- > \text{NO}_2^+$
16. The bent or V-shape of the molecule can be resulted from which of the following hybridization.
(A) sp^3 (B) sp^2 (C) Both (A) and (B) (D) None of these
17. When NH_3 is treated with HCl , in the product species H-N-H bond angle (in comparison to ammonia)
(A) Increases (B) Decreases (C) Remains same (D) Depends upon temperature
18. The correct order of bond angle is :
(A) $\text{H}_2\text{S} < \text{NH}_3 < \text{BF}_3 < \text{CH}_4$ (B) $\text{NH}_3 < \text{H}_2\text{S} < \text{CH}_4 < \text{BF}_3$
(C) $\text{H}_2\text{S} < \text{NH}_3 < \text{CH}_4 < \text{BF}_3$ (D) $\text{H}_2\text{S} < \text{CH}_4 < \text{NH}_3 < \text{BF}_3$
19. Which of the following statements concerning the molecule $\text{H}-\text{C}\equiv\text{C}-\text{CH}_3$ is / are not correct.
(A) It has 6s and 2p bonds
(B) Both the p-bonds are present in the same plane
(C) The central carbon atom is sp hybridised while the terminal atoms are sp^2 hybridised
(D) The molecule is linear
20. Which is/are in linear shape ?
(A) NO_2^+ (B) XeF_2 (C) I_3^- (D) I_3^+
21. Which is true about NH_2^- , NH_3 , NH_4^+ ?
(A) Hybridization of N is same. (B) No. of lone pair of electron on N are same.
(C) Molecular geometry (i.e. shape) is different. (D) Bond angle is same.
22. Identify the correct option(s)
(A) $\text{NH}_4^+ > \text{NH}_3 > \text{NH}_2^-$ order of bond angle
(B) $(\text{CH}_3)_3\text{B}$ is a trigonal planar molecule (not considering the H-atoms on 'C')
(C) In NH_4Cl 'N' atom is in sp^3d hybridisation
(D) In S_8 molecule a total of 16 electrons are left on all the 'S' atoms after bonding.
23. Which of the following molecule (s) has/have bond angle close to 90° ?
(A) NH_3 (B) H_2S (C) PH_3 (D) ICl_3

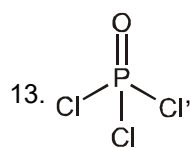
24. Match the Column I with Column II (Multiple Match Type)
- | Column – I | Column – II |
|--------------------|---|
| (A) SF_2 | (p) sp^3 and bent |
| (B) KrF_4 | (q) two lone pairs |
| (C) NOCl | (r) bond angle $< 109^\circ 28'$ |
| (D) NF_3 | (s) sp^2 and bent |
| | (t) sp^3d^2 and square planar |
25. The structure of IO_2F_2^- is analogous to :
 (A) SF_4 (B) XeO_2F_2 (C) F_2SeO_2 (D) (A) and (B) both
26. Write (i) number of bond pairs and lone pairs on the central atom (ii) the shape of the molecules (iii) hybridization of the central atom.
 (a) SF_4 (b) XeOF_4
27. Among the following species, identify the isostructural pairs : NF_3 , NO_3^- , BF_3 , H_3O^+ , HN_3 :-
 (A) $[\text{NF}_3, \text{NO}_3^-]$ and $[\text{BF}_3, \text{H}_3\text{O}^+]$ (B) $[\text{NF}_3, \text{HN}_3]$ and $[\text{NO}_3^-, \text{BF}_3]$
 (C) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{NO}_3^-, \text{BF}_3]$ (D) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{HN}_3, \text{BF}_3]$
28. Which of the following does not have same shape :-
 (A) $\text{SO}_4^{2-}, \text{S}_5^{2-}$ (B) $\text{ICl}_4^\ominus, \text{I}_5^\ominus$ (C) CO_2, SO_2 (D) $\text{NO}_2^\oplus, \text{NO}_2^\ominus$
29. Among the following orbital bonds, the angle is minimum between :-
 (A) sp^3 bonds (B) p_x and p_y orbitals
 (C) $\text{H}-\text{O}-\text{H}$ in water (D) sp bonds
30. In H_2O , the bond angle $\text{H}-\text{O}-\text{H}$ is $104^\circ 28'$ but in H_2S , H_2Se and H_2Te the bond angles are pretty close to 90° . This suggests that :-
 (A) oxygen uses sp^2 -hybrid orbitals while S, Se and Te use sp^3 -hybrid orbitals for bonding with the hydrogen atoms
 (B) oxygen uses sp^3 -hybrid orbitals to bond with the two hydrogen atoms while S, Se and Te use almost pure p orbitals
 (C) oxygen uses sp^3 -hybrid orbitals while S, Se and Te utilize d orbitals for bonding with the hydrogen atoms
 (D) all the atoms use pure p orbitals to bond with the hydrogen atoms
31. Which of the following statements is/are correct ?
 (A) NH_2^+ shows sp^2 – hybridisation whereas NH_2^- shows sp^3 – hybridisation
 (B) $\text{Al}(\text{OH})_4^-$ has a regular tetrahedral geometry
 (C) sp^2 -hybridized orbitals have equal s-and p-character
 (D) usually hybridized orbitals form s-bonds
32. Match the Column I with Column II (Multiple Match Type)
- | Column-I | Column-II |
|--------------------------------------|-------------------------|
| (A) $\text{B}_3\text{N}_3\text{H}_6$ | (p) planar geometry |
| (B) H_2O_2 | (q) non planar geometry |
| (C) B_2H_6 | (r) no lone pair |
| (D) I_2Cl_6 | (s) non polar molecule |
33. Statement-I : N_2O , CO_2 & I_3^- are isostructural.
 Because
 Statement-II : All three have same hybridise central atom.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

34. Among the following species, identify the isostructural pairs : $\text{NF}_3, \text{NO}_3^-, \text{BF}_3, \text{H}_3\text{O}^+, \text{OF}_2$
- (A) $[\text{NF}_3, \text{NO}_3^-]$ and $[\text{BF}_3, \text{H}_3\text{O}^+]$ (B) $[\text{NF}_3, \text{OF}_2]$ and $[\text{NO}_3^-, \text{BF}_3]$
 (C) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{NO}_3^-, \text{BF}_3]$ (D) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{OF}_2, \text{BF}_3]$
35. The hybridisation and shape of BrF_3 molecule are :
- (A) sp^3d and bent T shape (B) sp^2d^2 and tetragonal
 (C) sp^3d and bent (D) none of these
36. CO_2 is isostructural with
- (I) HgCl_2 (II) NO_2 (III) SnCl_4 (IV) C_2H_2
 (A) I and III (B) II and IV (C) I and IV (D) III and IV
37. The bond angle and hybridization in ether (CH_3OCH_3) is :
- (A) $106^\circ 51'$, sp^3 (B) $104^\circ 31'$, sp^3 (C) $> 109^\circ 28'$, sp^3 (D) None of these
38. Which of the following pairs is (are) isostructural?
- (A) SF_4 and SiF_4 (B) SF_6 and SiF_6^{2-} (C) SiF_6^{2-} and SeF_6^{2-} (D) XeO_6^{4-} and TeF_6^{2-}
39. Which of the following statements is/are correct ?
- (A) NH_2^+ shows sp^2 – hybridisation whereas NH_2^- shows sp^3 – hybridisation
 (B) $\text{Al}(\text{OH})_4^-$ has a regular tetrahedral geometry
 (C) sp^2 – hybridized orbitals bonded to same substituents have equal s- and p- character
 (D) Hybridized orbitals always form s - bonds
40. There is change in the hybridisation when:
- (A) NH_3 combines with H^+ (B) AlH_3 combines with H^-
 (C) NH_3 forms NH_2^- (D) SiF_4 forms SiF_6^{2-}
41. Shape of NH_3 is very similar to :
- (A) SeO_3^{2-} (B) CH_3^- (C) BH_3 (D) CH_3^+
42. Match the Column I with Column II (Multiple Match Type)
- | Column I | Column II |
|----------------------|--|
| (A) BrF_3 | (P) One angle $\neq 90^\circ$ |
| (B) TeF_5^- | (Q) Central atom is sp^3d^2 hybridised |
| (C) IF_7 | (R) Non planar |
| (D) XeF_4 | (S) Polar |
43. In which pair of molecules bond angles are not same:
- (A) CCl_4 & SiCl_4 (B) NH_4^+ & NF_4^+ (C) ClF_6^+ & SF_6 (D) None
44. The molecules / ions which are planar as well as polar.
- (A) BF_3 , H_2O , HF , NH_2^- (B) SnCl_2 , I_3^+ , NH_2^- , IF_3
 (C) CO_3^{2-} , I_3^- , SF_2 , XeF_2 (D) NO_2^- , XeF_4 , ICl_4^- , NH_2^+
45. Statement-1 : On decreasing s-character in hybrid orbitals, bond angle decreases.
 Statement-2 : p-orbitals are at 90° to one another.
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

46. Which of the following should have pyramidal shape :
 (A) $[\text{ClOF}_2]^+$ (B) ICl_3 (C) $[\text{BrCl}]^-$ (D) All of these
47. Which of the following statements is false for XeO_3F_2 ?
 (A) Hybridisation of central atom xenon is sp^3d . (B) The compound is non-polar.
 (C) The compound has pp-dp bonding. (D) None.
48. Amongst NO_3^- , AsO_3^{3-} , CO_3^{2-} , ClO_3^- , SO_3^{2-} and BO_3^{3-} , the non-planar species are :-
 (A) CO_3^{2-} , SO_3^{2-} and BO_3^{3-} (B) AsO_3^{3-} , ClO_3^- and SO_3^{2-}
 (C) NO_3^- , CO_3^{2-} and BO_3^{3-} (D) SO_3^{2-} , NO_3^- and BO_3^{3-}

Answer key

1. D 2. D 3. A 4. B 5. D 6. B 7. A
 8. A 9. D 10. B 11. A 12. D



double bond occupies large area and has large electron density. So there is intrinsic repulsion between $\text{P}=\text{O}$ and $\text{P}-\text{Cl}$ bond pairs. To minimize this repulsion bond angle decrease from 109.5° to 103.5° .

14. (i) $\text{NO}_2^+ > \text{CO}_3^{2-} > \text{BF}_4^-$ (ii) $\text{SO}_3 > \text{SO}_2\text{Cl}_2 > \text{SF}_6$ (iii) $\text{OBr}_2 > \text{OCl}_2 > \text{OF}_2$ (iv) $\text{PCl}_5 = \text{PBr}_5 > \text{XeO}_2\text{F}_2$
15. B 16. C 17. A 18. C 19. B,C 20. A,B,C
 21. A,C 22. A,B 23. B,C,D 24. (A - p, q, r) ; (B - q, r, t) ; (C - s) ; (D - r)
 25. D
 26. (a) SF_4 : BP = 4, LP = 1, sp^3d , see-saw (b) XeOF_4 : BP = 5, LP = 1, sp^3d^2 , square pyramidal
 27. C 28. A,B,C,D 29. B 30. B
 31. C 32. A - (p, r, s), B - (q), C - (q, r, s), D - (p, s) 33. C 34. C
 35. A 36. C 37. C 38. B 39. A,B,C 40. B,D
 41. A,B 42. (A) P,S (B) P,Q,R,S (C) P,R (D) P,Q 43. D 44. B 45. B
 46. B 47. D 48. B